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Ontology Mapping and Data Processing of the ITHANET Database

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1. Ontology Mapping

One of the main tasks of the internship was the ontology mapping of the terminology used in the ITHANET databases. This work formed part of the process of *FAIRifying* the data, following the FAIR principles, according to which data should be Findable, Accessible, Interoperable, and Reusable. Associating database terminology with established biomedical ontologies contributes to interoperability by providing standardised and machine-readable representations of the data. It can also facilitate the interpretation and reuse of the data by researchers, databases, and computational tools.

The ontology mapping was performed on data extracted from the ithabase_live and ithaphen_live databases. Since the databases contain multiple tables with different types of information, the mappings were organised into separate CSV files according to the database, table, and type of information being mapped.

1.1. Organisation of the Mapping Files

The names of the mapped files have the following format:

`<database_name>_<table_name>_<labels_or_values>_completed.csv`

where:

- database_name corresponds to either ithabase_live or ithaphen_live,
- table_name corresponds to the name of the database table from which the information was extracted,
- labels refers to the names of the fields or columns of the table and
- values refers to the values stored within the fields of the table.

1.2. Mapping of the table labels

The labels of each table were turned from a row into a column and then two columns were added next to them with the following fields:

- the preferred name of the corresponding ontology and
- the URL of the ontology.

The ontology preferred name was stored in a separate column because the terminology used within the database did not always exactly match the preferred terminology used by the corresponding ontology.

1.3. Identification and Selection of Ontology Terms

Different biomedical ontologies were investigated depending on the type of term being mapped. Among the resources considered were ontologies such as NCIT, SNOMED CT, MONDO, HPO, EFO, and others available through biomedical ontology repositories, with a preference towards SNOMED whenever possible.

1.4. Merging the Ontologies

Once the ontology mappings had been completed across the different database tables, a Python script

`Merge_Value_Ontologies.py`

was developed to combine the mapped values from the different CSV files into a single dataset.

The script extracts the terms and their associated ontology information from the completed files and combines them into a common output file. If the same original term is associated with more than one ontology concept, the different ontology mappings are preserved.

1.5. Outcome of the Ontology Mapping

The ontology mapping process resulted in a collection of completed mapping files covering both database labels and relevant database values, together with a merged dataset containing ontology mappings collected from across the different tables.

A particularly useful tool during this work was Ontotext Refine. It helped organise and categorise large numbers of terms, making the ontology mapping process more manageable. Features such as text facets were especially useful for grouping similar values, identifying repeated terms, and quickly checking how often certain terms appeared in the dataset. Ontotext Refine also allowed multiple entries to be modified at the same time, which made corrections and standardisation much faster than editing each value individually. Its search and filtering functions were also useful for locating specific terms and checking the consistency of the mappings across the dataset.

2. Data Processing

The second part of the internship focused on the processing of data from the IthaGenes database for submission to ClinVar. Most of this work was carried out by modifying and expanding the existing R Markdown script *export-ithagenes-to-ClinVar.Rmd*. The main objectives of the modifications were to prevent the loss of potentially relevant variants, preserve the results of previous runs, facilitate comparisons between successive exports, and make changes in the data easier to identify and verify.

2.1. Overview of the Export Pipeline

The first part of the script establishes a connection to the *ithabase_live* database and extracts the information required for the ClinVar submission. Data from several related IthaGenes tables are combined to obtain information concerning the variants, their HGVS descriptions, genes, classifications, phenotypes, references, and other information required during the subsequent processing steps.

The extracted data are then processed to prepare the variants for validation and eventual ClinVar submission. The HGVS name is particularly important in this process because it provides a standardised description of genetic variants. VariantValidator is used within the workflow to validate and standardise the variant descriptions before the final submission files are generated. The final data are organised into five main thalassemia submission groups:

- β^0 thalassemia,
- β^+ thalassemia,

- α^0 thalassemia,
- neutral/benign variants, and
- thalassemia Dominant

Separate ClinVar submission files are generated for these groups.

Variants that do not belong to one of the four candidate submission groups are not discarded. Instead, the remaining IthaGenes IDs are grouped according to their associated phenotype and saved separately in:

```
outputs/ithaids_not_in_candidates_by_phenotype/
```

The files within this folder therefore contain records that were extracted from IthaGenes but were not included among the candidates for the four principal submission groups. A dated intermediate file containing the candidates before VariantValidator processing is also retained:

```
candidates_before_variantvalidator_DATE.csv
```

This file provides a record of the variants selected for the four main submission groups before validation. Together with the phenotype-specific files containing the IthaGenes IDs that were not selected as candidates, these outputs provide a more complete account of how the original extracted dataset is divided during the processing pipeline.

Another important modification concerned the management of the files produced by the script. Previously, the export pipeline was overwriting files produced by earlier executions. This made it difficult to reconstruct previous results or compare different versions of the export. The output system was therefore modified so that newly generated files are stored inside an *outputs* directory and include the date of execution in their filenames. Instead of replacing an existing file, each run produces a new dated version, allowing the results of previous executions to be retained.

2.2. Modification of the Variant Validation Workflow

One of the most important modifications was the order in which variants were categorised and validated. In the original workflow, the Batch VariantValidator step was performed before the variants were separated into their corresponding thalassemia groups. Variants for which a suitable validated HGVS description could not be obtained were therefore excluded at an early stage.

This was problematic because failure to obtain a match during validation did not necessarily mean that the original database record should be discarded. For example, an HGVS name may have been modified or updated and therefore no longer correspond exactly to the form present in the reference data. Consequently, potentially relevant records could disappear before the final submission groups were created.

The workflow was modified so that the candidate variants are first selected and separated into their corresponding categories directly from the IthaGenes data. Variant validation is then performed afterwards. The VariantValidator results are joined back to the complete set of candidate variants rather than being used to determine which variants are retained.

As a result, the four thalassemia files now preserve all candidate records, including variants for which VariantValidator does not return a corresponding validated HGVS name. For these records, the relevant validated fields can remain empty while the original variant and its associated information are retained. This makes it possible to distinguish a validation problem from the

absence of the original database record.

A dated intermediate file containing the candidates before VariantValidator processing is also retained:

`candidates_before_variantvalidator_DATE.csv`

This file provides a record of the variants selected before validation and can therefore be used to verify whether records have been lost or modified during the subsequent stages of the pipeline.

2.3. Comparison with the Previous ClinVar Submission

An additional part of the R code was developed to compare the latest results of the export pipeline with the previous ClinVar submission. The four newly generated thalassemia datasets are first combined and compared with the previous ClinVar submitter report. Records from the two datasets are linked using their local IthaGenes identifiers. In the previous ClinVar report, these identifiers are represented by the *Your_ variant_ id* field, while the newly generated files use the corresponding IthaGenes identifier.

A new R chunk was added to merge the latest generated data with the previous submission while preserving the columns from both sources. The resulting file is written as:

`merged_clinvar_submission_with_report_DATE.csv`

This merged dataset provides a complete side-by-side representation of the previous submission and the current export. It is particularly useful for manually inspecting individual variants and determining why a particular record has changed.

2.4. Identification of Modified Variants

The comparison between the previous ClinVar submission and the latest export focuses particularly on three fields:

- HGVS name,
- gene, and
- clinical significance.

For the current export, the HGVS description is derived from the validated variant information produced during the VariantValidator processing. This is compared with the variant description contained in the previous ClinVar report.

To make these differences easier to identify, another R chunk was added to generate three Boolean columns. Each column indicates whether one of the three monitored fields has changed between the previous and current versions. The values are stored as *TRUE* or *FALSE*, depending on whether a difference is detected. This makes it possible to distinguish, for example, a variant for which only the HGVS name has changed from one for which the gene or clinical significance has also been modified. The resulting comparison can therefore be filtered according to the type of modification and provides a rapid method for reviewing potentially important changes before submission. A dedicated file containing the modified submissions is also produced:

`modified_clinvar_submissions_DATE.csv`

This file provides a more focused view of the records for which differences were identified.

2.5. Detailed Tracking of Submission Changes

In addition to the merged and Boolean comparison tables, a separate output was introduced to record the specific differences between the previous ClinVar submission and the current export. This file is written in the form:

`clinvar_submission_changes_DATE.csv`

The purpose of this file is to provide a direct record of changes affecting the HGVS name, gene, and/or clinical significance. While the Boolean columns indicate whether a particular property has changed, this output allows the corresponding modifications to be examined more directly. Together, the merged comparison file, the Boolean indicators, and the detailed changes file provide different levels of information for quality control. The merged file contains the complete information from the old and new datasets, the Boolean columns make changed records easy to identify and filter, and the changes file provides a focused record of the specific modifications detected between the two versions.

2.6. Quality Control of Excluded and Unmatched Variants

Particular attention was given to variants that could previously have been lost during the validation process. Following the modification of the workflow, the four thalassemia datasets retain the complete set of selected candidates even when a validated HGVS name is not obtained. This ensures that variants are not automatically excluded solely because VariantValidator does not return a validated HGVS description.

Before VariantValidator is applied, the complete set of selected candidate variants is exported as:

`candidates_before_variantvalidator_DATE.csv`

This file provides a record of the variants entering the validation stage and can be compared with the datasets generated later in the workflow. In this way, the processing of individual variants can be followed throughout the pipeline and variants for which validation was unsuccessful can still be retained for further inspection rather than being silently removed.

An important distinction must be made for variants whose HGVS names are not found in the reference 1910 file. These records can represent three different situations: the variant may have an HGVS description that has subsequently been modified, or an HGVS description may be unavailable altogether (i.e. ithaID:400, ithaID:410, ithaID:316, ithaID:299, ithaID: 332, ithaID:334, ithaID:337, ithaID:348), or the variant was not in the database when variant validator was last run. Keeping these records in the four thalassemia datasets allows these three cases to be investigated separately rather than automatically removing both during validation.

The ithaIDs with no assigned HGVS name can be found in

`qc_no_hgvs_in_ithagenes_DATE.csv`

Following the changes to the validation workflow, the previously created comparison tables were also checked to ensure that the Boolean indicators for HGVS name, gene, and clinical significance continued to represent the correct differences between the previous submission and the latest export.

2.7. Verification of ORPHANET Identifiers

An additional quality-control step was implemented for the *neutral_ benign* submission group to ensure that the disease identifiers included in the ClinVar submission were consistent with the molecular description of each variant. The appropriate thalassemia category is determined automatically from the HGVS description and the associated globin gene.

For each variant, the HGVS description returned by VariantValidator is used preferentially. When no validated HGVS description is available, the HGVS annotation stored in IthaGenes is used instead. The gene symbol provided by VariantValidator is retained when available otherwise, the gene is inferred from the HGVS description. Variants associated with HBA1 or HBA2 are classified as belonging to the alpha-globin locus, whereas variants associated with HBB are classified as belonging to the beta-globin locus. Genomic HGVS descriptions are also considered, allowing variants to be assigned to the appropriate locus even when a specific gene cannot be identified directly.

Based on this classification, the corresponding ORPHANET identifier is assigned automatically in the final ClinVar submission file. Variants belonging to the alpha-globin locus are assigned *ORPHA:846*, corresponding to alpha-thalassemia, while variants belonging to the beta-globin locus are assigned *ORPHA:848*, corresponding to beta-thalassemia.

As a final verification step, records for which either the condition identifier type or the ORPHANET identifier remains empty are extracted into a separate quality-control file. These records can then be reviewed manually before submission. This procedure ensures greater consistency between the molecular description of each variant and the disease information included in the ClinVar submission and reduces the risk of assigning an incorrect thalassemia category.

3. Conclusion

This internship focused on two aspects of biomedical data management: the ontology mapping of terminology used in the ITHANET databases and the improvement of the IthaGenes data-processing pipeline for ClinVar submission. The ontology mapping work contributed to the FAIRification of the data by linking database terminology to standardised biomedical ontology concepts, thereby improving its interoperability and potential reuse. The mappings were organised systematically across the different database tables, and a Python script was developed to merge the completed mappings into a common dataset while preserving multiple ontology associations when necessary.

The second part of the internship focused on improving the reliability and traceability of the ClinVar export workflow. The processing pipeline was modified so that candidate variants are retained even when VariantValidator does not return a validated HGVS description, preventing potentially relevant records from being removed prematurely. The output system was also reorganised to preserve dated versions of generated files, making successive runs easier to compare and review. Additional quality-control outputs were introduced to identify changes in HGVS descriptions, genes, and clinical significance between the previous ClinVar submission and the latest export, while variants excluded from the main submission groups were retained separately according to phenotype.